Goal & Scope Definition

CIVE475 Summer 2025 Life Cycle Assessment

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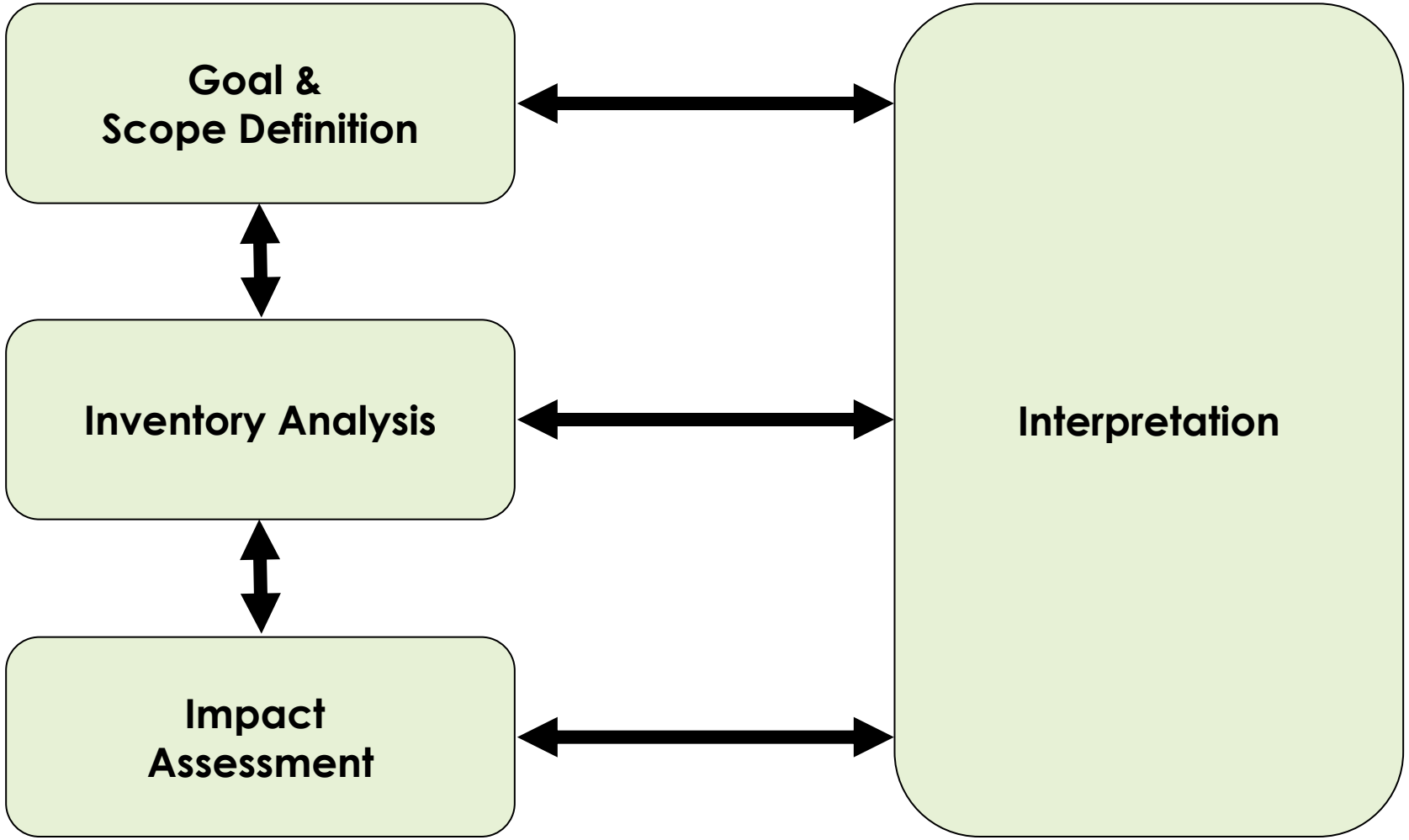
Objectives of this unit

- Introduce the first component of the ISO LCA framework: Goal & Scope Definition
- Understand the aspects included within goal & scope definition
- Describe Study Design Parameters (SDPs)
- Be able to write a goal & scope definition for an LCA of a product

LCA Framework



ISO 14040: Goal & Scope Definition

- ✓Statements of intent for the LCA study
- ✓**Study Design Parameters**
- ✓Explicitly state reason for the study

Goal

The **goal** of the study should state:

- ✓ Intended application
- ✓ Reasons for carrying out study
- ✓ Intended audience
- ✓ Whether results intended to be used in comparative assertions disclosed to the public

"Who might care about this and why?"
"Why did we do it and what will we do with it?"

Examples of intended applications

- ✓ Comparison of specific products or services
- ✓ Benchmarking vs. product group average
- ✓ Environmental Product Declarations
- ✓ Policy development and enforcement
- ✓ Identification of key environmental performance indicators of product
- ✓ Green procurement
- ✓ Ecodesign
- ✓ Carbon footprint
- ✓ Greening supply chain
- ✓ Monitoring environmental impacts
- ✓ Corporate environmental reporting
- ✓ …

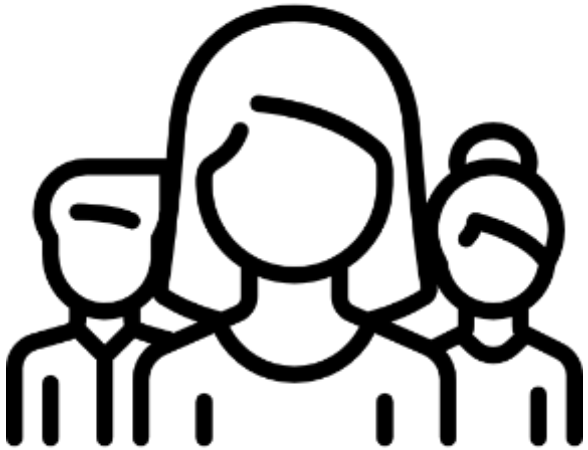
Defining the goal of an LCA is an iterative process

We want to do an LCA!

Why?

…

…



Commissioner

Practitioner

Main types of LCA to commission

	Single Product	Comparative
External Communications	**Critically reviewed LCA** • ISO 14044 Single Product LCA • EPD • ISO 14067 (Carbon Footprint) or 14046 (Water Footprint) • Only one reviewer needed	**Critically reviewed Comparative LCA** • ISO 14044 Comparative LCA • ISO 14067 (Carbon Footprint) • Include extensive sensitivity and uncertainty analyses • Three reviewers needed
Internal Communications	**Screening LCA** • Hotspot analysis • High-level assessment with literature review • Assess scope for public facing study	**Screening Comparative LCA** • Evaluate advantage • Identify most promising applications/conditions • Assess scope for public facing study

Example: LCA of artificial vs natural Christmas trees

Intended application (and audience) ✓ "The findings of the study are **intended to be used … educated external communication** and **marketing aimed at the American Christmas tree consumer**."

Reasons ✓ "The goal …is **to understand the environmental impacts** of .. the most common artificial Christmas tree and the most common natural Christmas tree, and to analyze how their environmental impacts **compare**."

Audience (and reasons) ✓ "This comparative study **is expected to** be released to the **public** by the American Christmas Tree Association **to refute myths and misconceptions about the relative difference in environmental impact by real and artificial trees**."

Case Study: Goal & Scope

Report No. CSS18-10
September 14, 2018

Beyond Meat's Beyond Burger Life Cycle Assessment:
A detailed comparison between a plant-based and an animal-based protein source

Martin C. Heller and Gregory A. Keoleian

Link

1. Introduction and Goal of the study

Beyond Meat has commissioned the Center for Sustainable Systems to conduct a life cycle assessment (LCA) of their Beyond Burger (BB) and compare it against a typical American ground beef patty. The primary reason for the study is to advance knowledge on the environmental impact of plant-based protein alternatives. In addition, Beyond Meat is interested in sharing results on the potential environmental benefits of BB publically to consumers and provide scientifically based evidence to support claims of the environmental impacts of consuming BB versus beef. A secondary goal is to provide Beyond Meat with a benchmark against which to measure future improvements in the environmental performance of the BB product chain as well as to highlight hotspots within the product chain. The impact categories of interest include greenhouse gas emissions, cumulative energy demand, water use, and land use.

The intended audience is both internal stakeholders at Beyond Meat, as well as external customers, consumers, and interested stakeholders.

A goal of the study is to conduct a comparative assessment of BB and beef and support comparative assertions intended for public communication. Accordingly, Critical review was conducted per Section 6.3 of the ISO 14044-2006 Standard. The ISO standard requires LCA studies to undergo a Critical Review by a panel of no less than three (3) reviewers when the results are intended to support comparative assertions that are intended to be disclosed to the public.

= Reason for carrying out the study

= Target audiences

= Intended application

Scope: ISO 14040

- ✓ **Product system(s)** to be studied
- ✓ **Functions of product system(s)**
- ✓ **Functional unit**
- ✓ **System boundary**
- ✓ **Allocation procedures**
- ✓ **Impact categories selected and methodology of impact assessment**, and subsequent interpretation to be used
- ✓ **Data requirements**
- ✓ **Assumptions**
- ✓ **Limitations**
- ✓ Initial **data quality** requirements
- ✓ Type of **critical review** (if any)
- ✓ Type and format of report required for the study

Scope: Study Design Parameters (SDPs)

- ✓ Series of parameters that **need to be qualitatively and quantitatively** described in an LCA according to ISO
- ✓ SDPs are a **summary** of the most important organizing aspects of an LCA
- ✓ A subset of required elements of an LCA under ISO
- ✓ Answer: **What did the study aim to do** (and not do)?

Functional unit

- ✓ "*Quantified performance of a product system for use as a reference unit*" **(ISO)**
- ✓ **Quantifies the function**
  - ✓ Describes primary function(s) of product system
  - ✓ Quantitative basis for which inputs and outputs relate
  - ✓ Unit for comparisons within LCA
- ✓ For some cradle-to-gate analysis, the function of the product could be unknown, and a **declared unit** is defined instead (ISO 21930)

Functional unit

- ✓ Must be **quantitative** and should have a '**unit**'
  - ✓ Unit does not have to be SI-type units
  - ✓ E.g., 1 pair of dry hands

- ✓ Should **relate to the function(s)** of product **rather than physical product**.
  - ✓ *E.g., "seating support for 1 person working at a computer for 1 year" rather than "1 computer workstation chair"*

Examples of functional units

Product/Service	Function	Functional unit
Car		
Truck		
Power plant		
Hand dryer		
Light bulb		
Cup of coffee		

Case studies of functional units

"one 4 oz. (¼ pound, 0.113 kg), uncooked burger patty delivered to retail outlets"

Link

"Packaging and distribution of 1000 litres of wine"

Link

"Cover 1 m^2 of wall while keeping color quality for 10 years"

Link

Case studies of functional units

"Offering the efficient protection of persons against malaria with a long-lasting mosquito net during a period of time encompassing 30 washes"

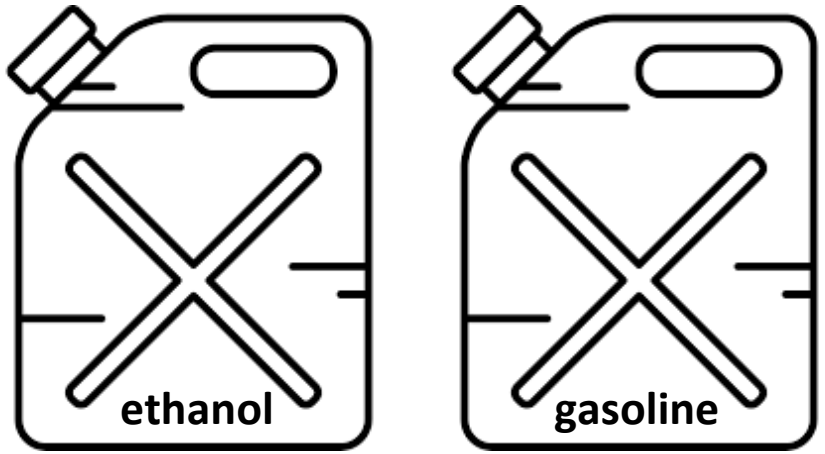
Link

DAIRY FARMERS OF CANADA QUALITY MILK

"one kilogram of fat- and protein-corrected milk (FPCM) produced at a Canadian farm and transported to a processing facility."

Link

Functional unit for comparisons



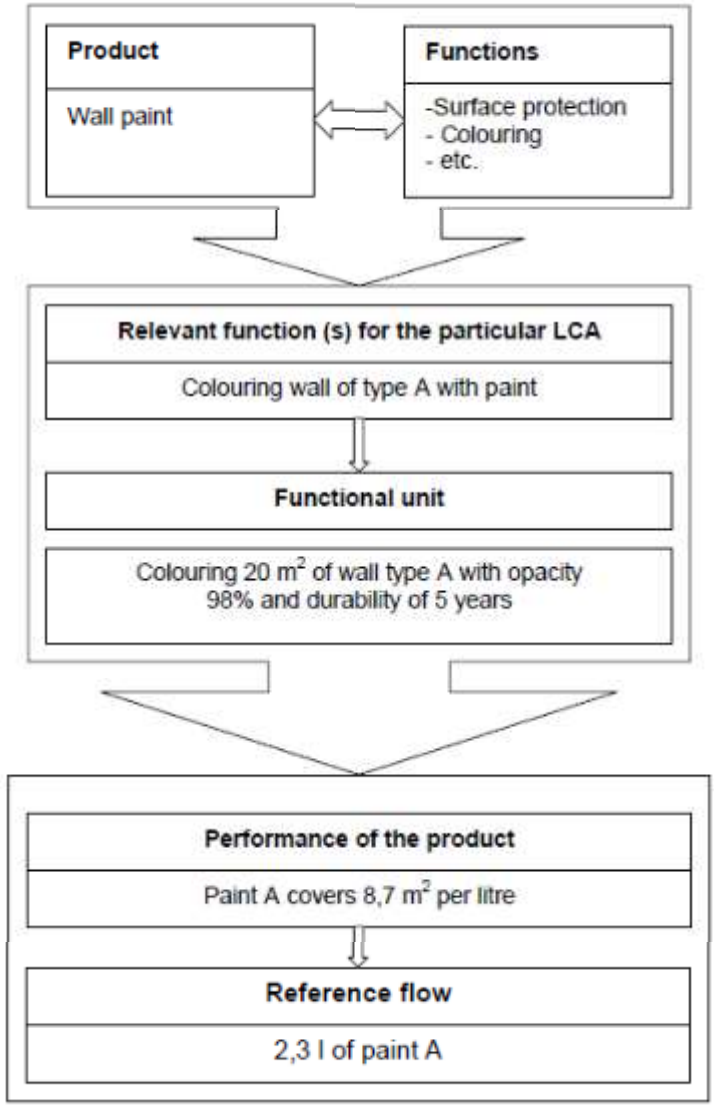
- ✓ **Consistent** across the alternatives
- ✓ Misleading to compare 1 liter of ethanol and 1 liter of gasoline
  - ✓ Ethanol approximately 2/3 of energy content on MJ/L basis
  - ✓ Functional unit = 1 MJ fuel or
  - ✓ Functional unit = 1 vehicle km travelled

Reference flow

- ✓ "*Measure of the outputs from processes in a given product system required to fulfil the function expressed by the functional unit*" **(ISO)**

- ✓ Having defined the functional unit, the reference flow is the **amount of product necessary to fulfil the function**

Functional unit and reference flow



Identify functions

Select relevant functions for LCA

Define functional unit

Identify product performance

Determine reference flow

ISO 14049

Example of functional unit and reference flow

- ✓ **Identify product systems**
  - ✓ Paper towel
  - ✓ Air-dryer
- ✓ **Identify relevant function for LCA**
  - ✓ Drying hands
- ✓ **Functional unit**
  - ✓ 1 pair of dried hands
- ✓ **Reference flows**
  - ✓ Average mass of paper required
  - ✓ Average volume of air required

Sensitivity analysis of functional unit

Journal of Cleaner Production 140 (2017) 470–478

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Protein quality as functional unit – A methodological framework for inclusion in life cycle assessment of food

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[c] Swedish University of Agricultural Sciences (SLU), Dept. of Food Science, P.O. Box 7051, 75007 Uppsala, Sweden

Sensitivity analysis of functional unit

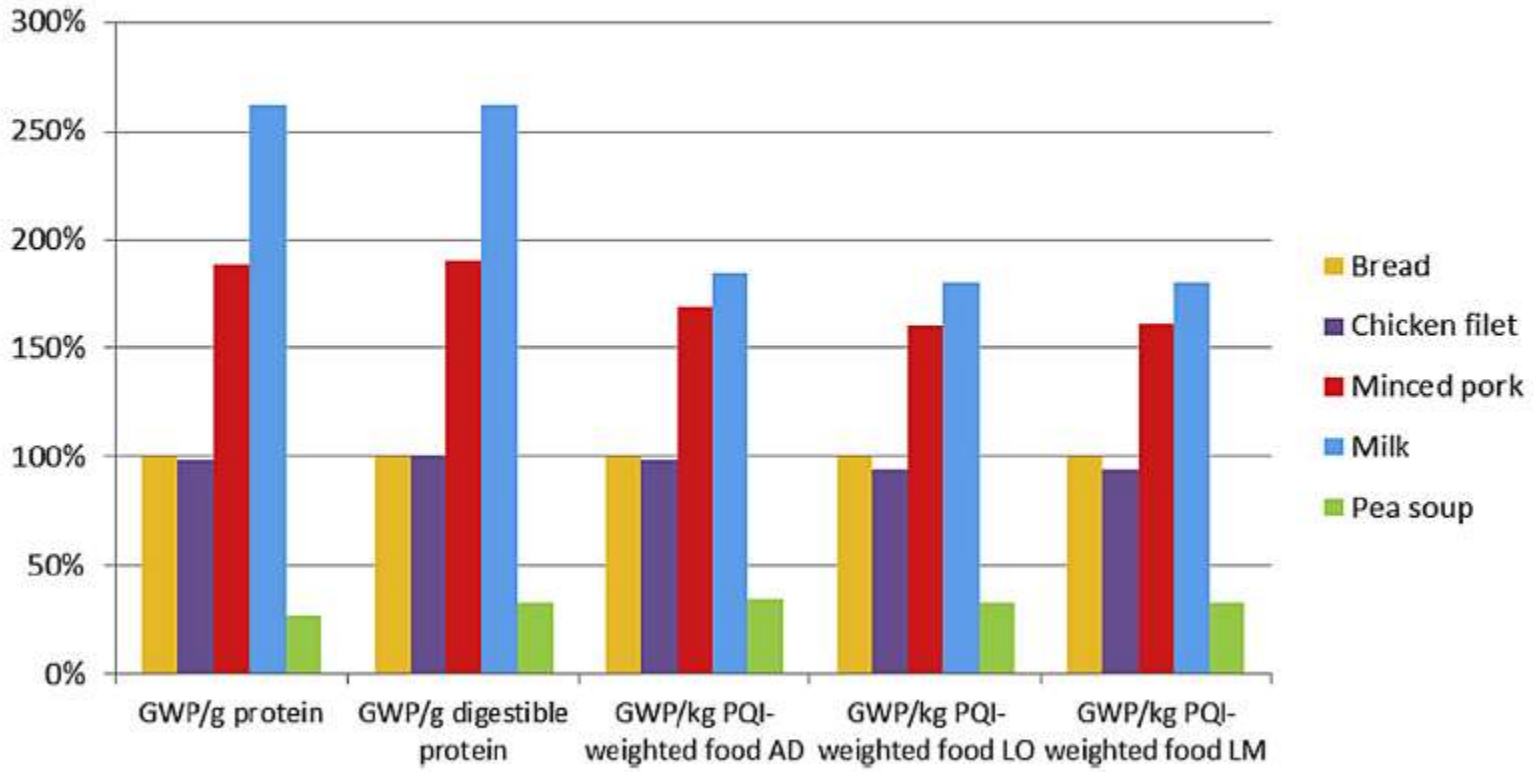


Fig. 2. Global warming potential (GWP, IPCC, 2014) (kg CO_2-equivalents) expressed with the three different functional units (g protein, g digestible protein, kg PQI-weighted food) the latter for three dietary contexts (AD = average Swedish diet, LO = lacto-ovo diet and LM = low-meat diet). Bread = 100%.

Product system: Terminology

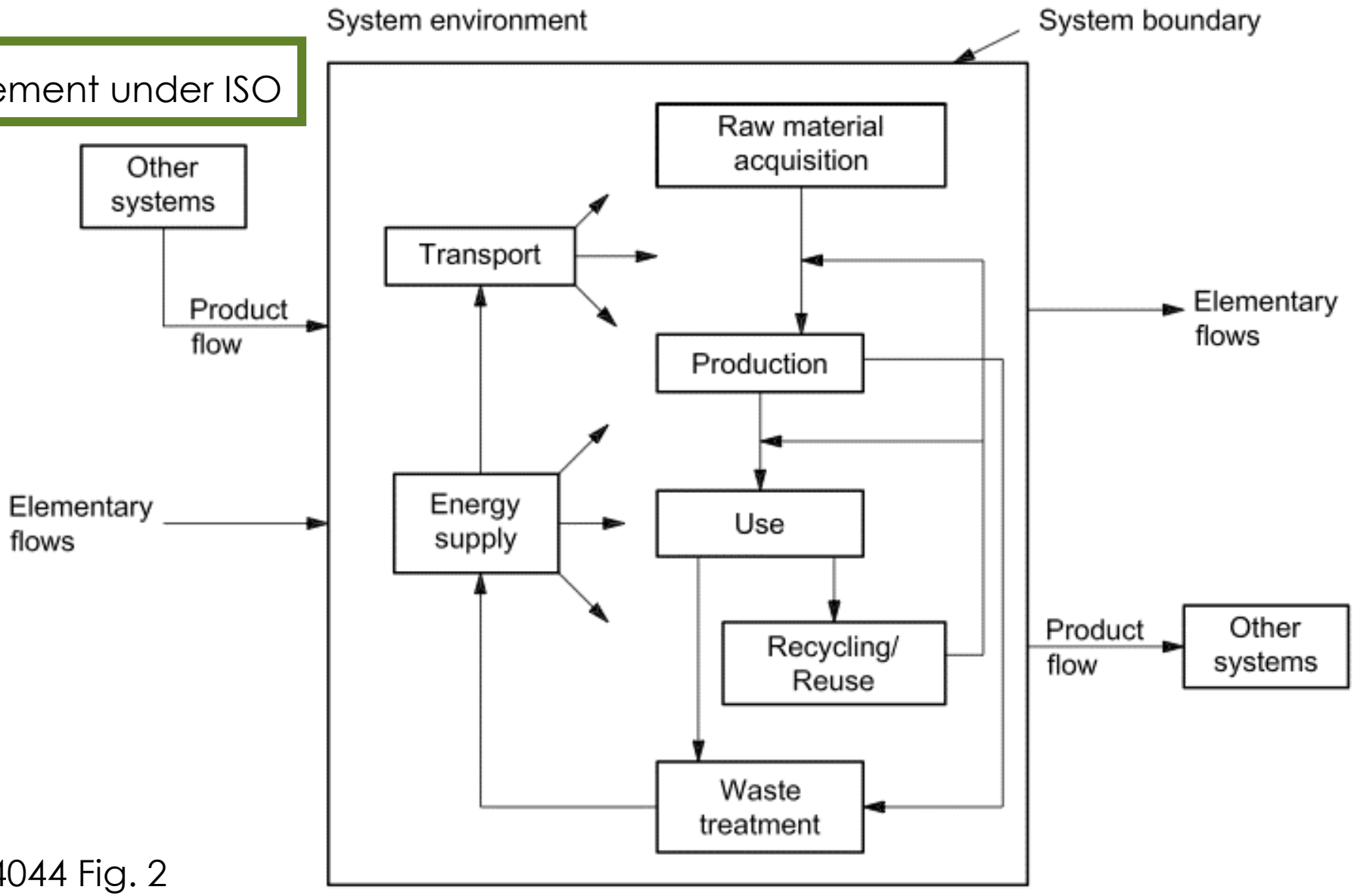
- ✓ Collection of **unit processes** with **elementary** and **product flows**, performing 1 or more **functions**, and which models life cycle of product

- ✓ **Unit process**: smallest element in life cycle inventory analysis for which input and output data quantified

- ✓ **Processes**: activities that transform inputs into outputs

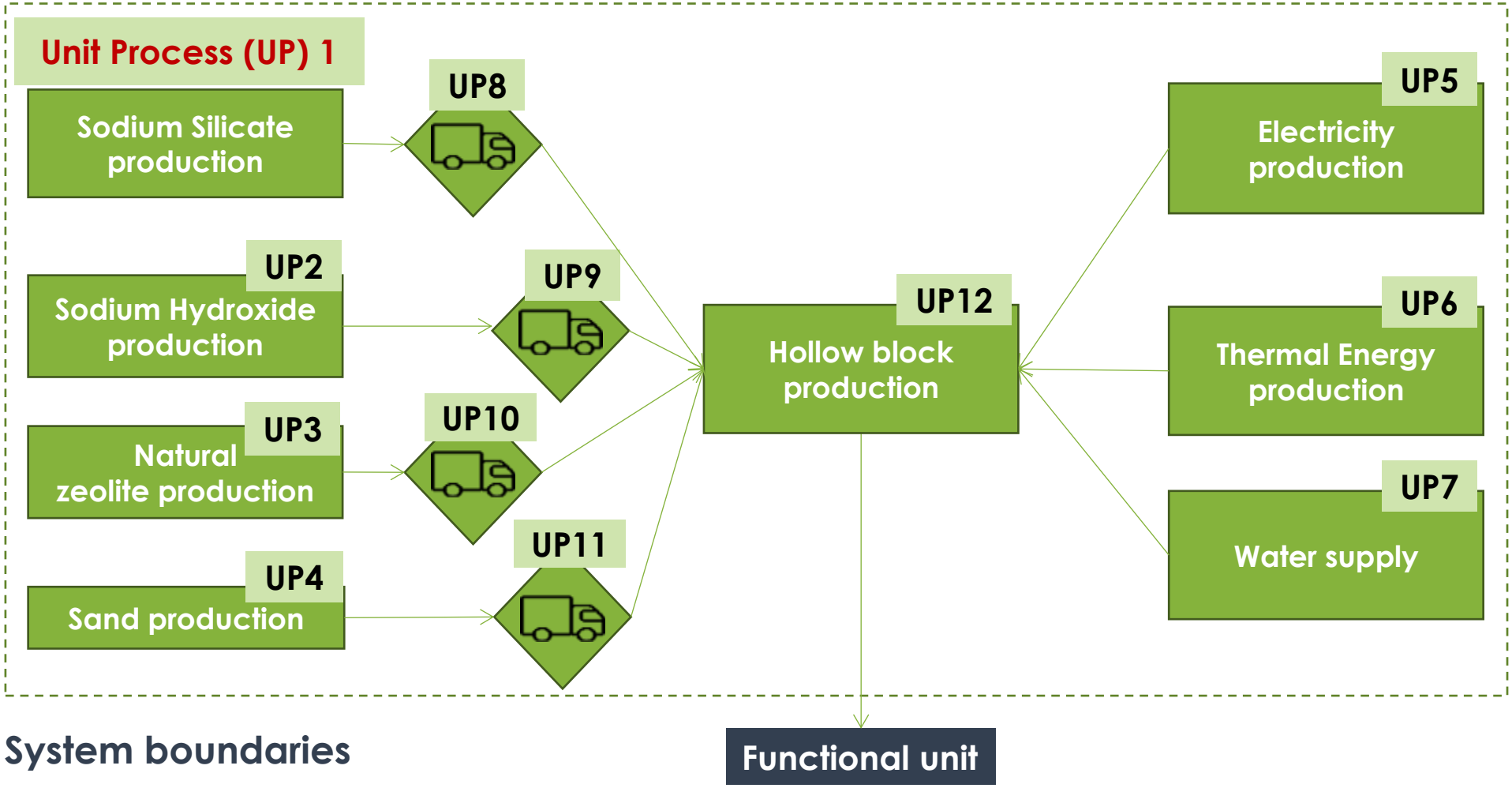
Product system diagram (simplified)

Not a requirement under ISO

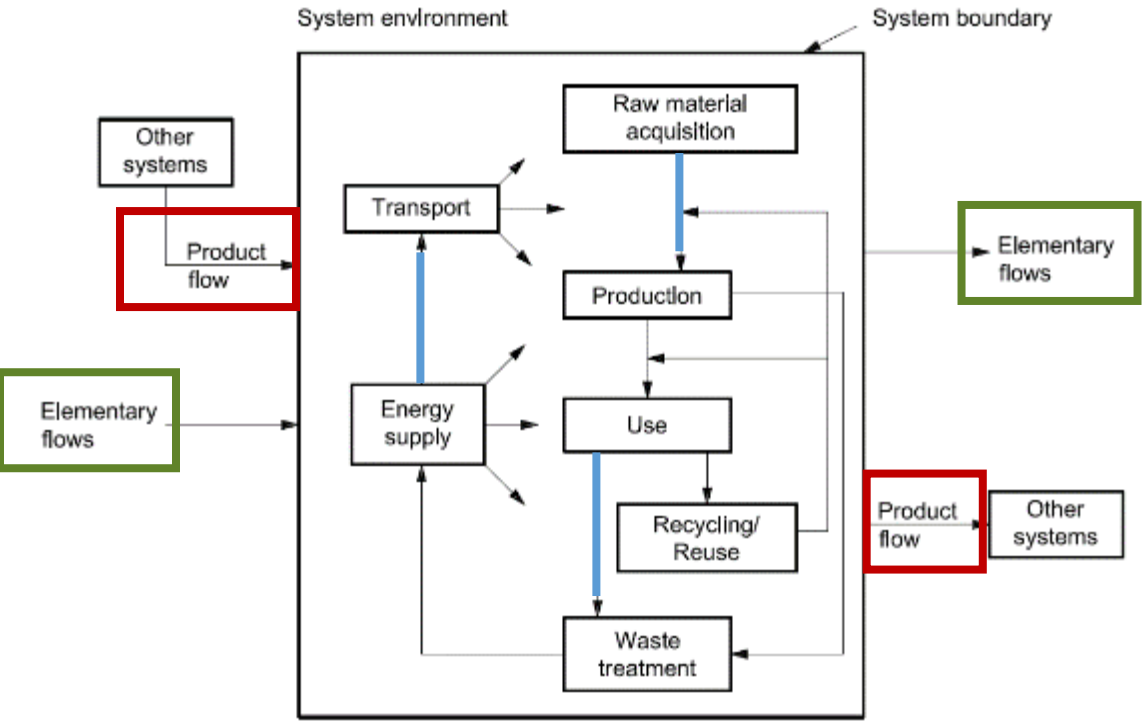


Source: ISO 14044 Fig. 2

Example Product system Geopolymer concrete

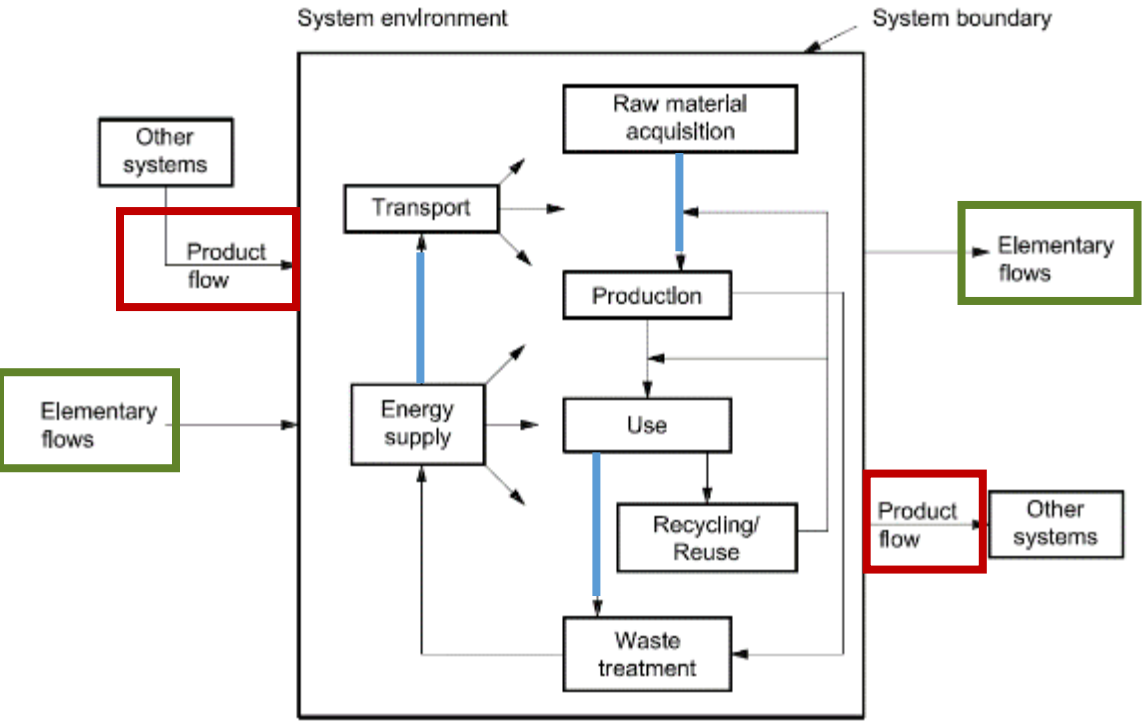


Product systems



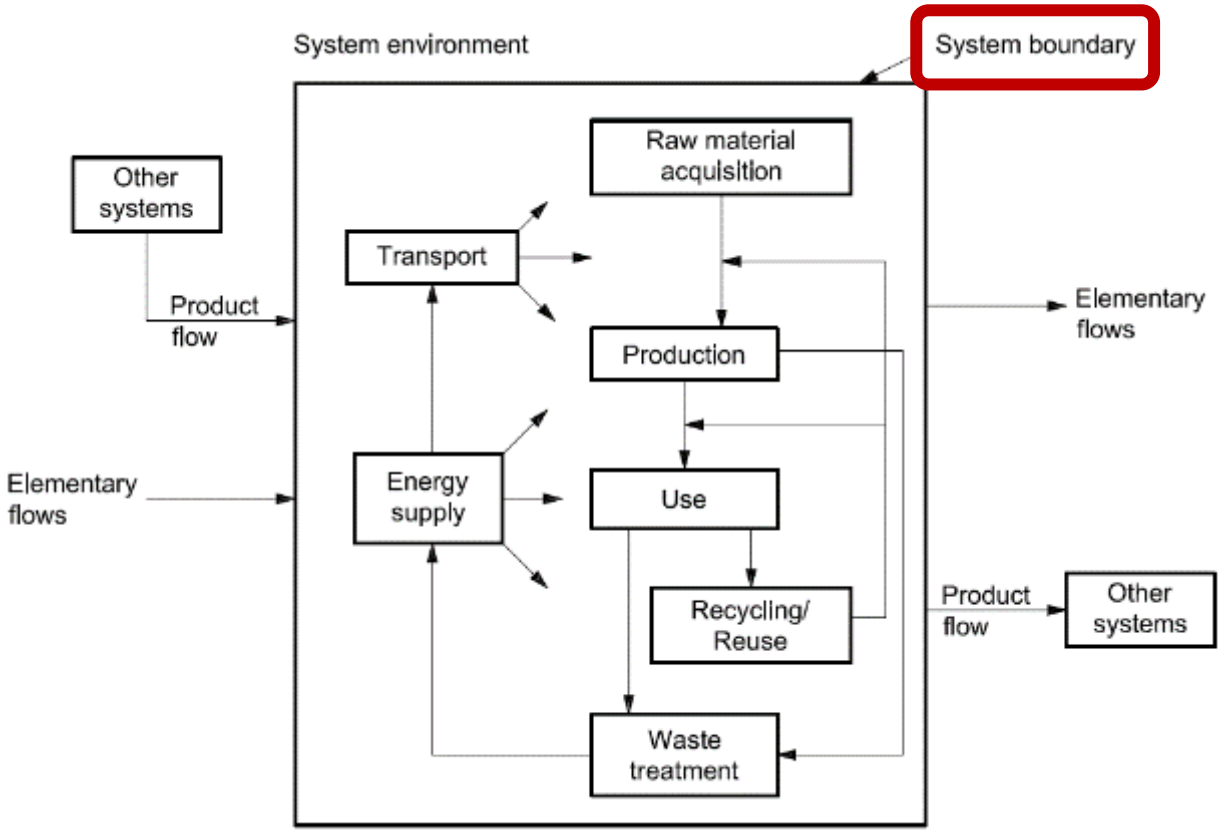
- ✓ **Elementary flows:** material/energy entering product system from the environment **without previous human transformation**, or leaving system and released into environment **without subsequent human transformation**
- ✓ **Intermediate flows:** material/energy flow occurring between unit processes of product system
- ✓ **Product flows:** products entering from or leaving to another product system

Product systems: Flow examples



- ✓ **Elementary flows**
  - ✓ **Entering unit process:** crude oil from ground, solar radiation
  - ✓ **Leaving unit process**: emissions to air, water, soil and radiation
- ✓ **Intermediate product flows**
  - ✓ Basic materials and subassemblies such as diesel fuel for transport
- ✓ **Product flows**
  - ✓ Recycled materials, components for reuse such as recycled steel

System boundaries

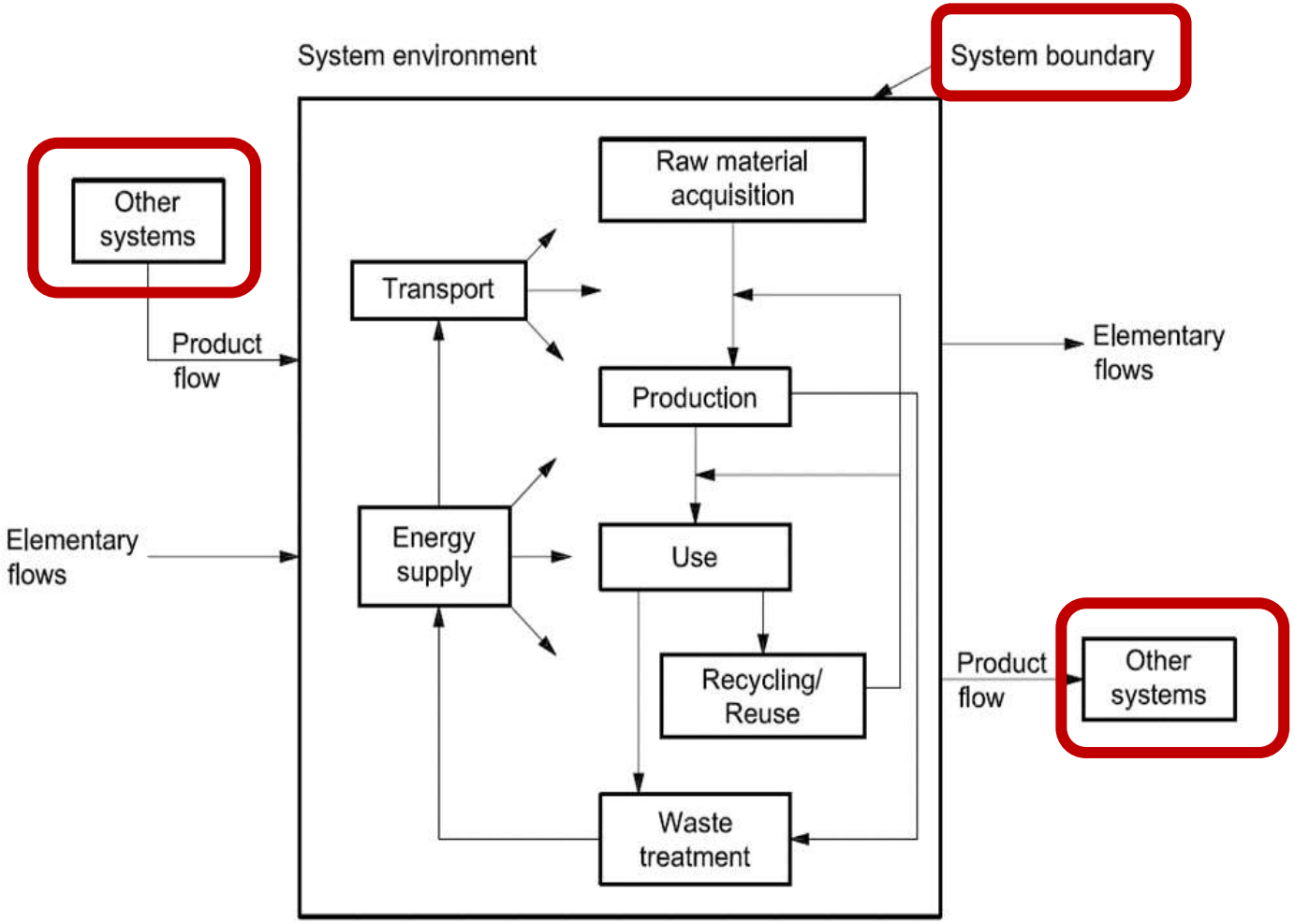


- ✓ **Indicates subset of overall processes and flows of product system included in study**
- ✓ Critical for understanding complexity of product system

System boundaries

- ✓ Justification for system boundary choices may include;
  - ✓ Statements about a process being assumed/found to have **negligible impact**
  - ✓ In comparative studies, if **identical processes exist in both product systems** and thus would not affect comparison
- ✓ Decisions to **omit** life cycle stages, processes, inputs or outputs must be clearly stated, and reasons and implications for omission explained
- ✓ **Cut-off** criteria

Cut-off criteria



- When aspect is considered to be negligible or irrelevant
- When only certain life cycle stages are included
  - E.g., cradle to gate
- Due to lack of resources
- ISO focuses on mass, energy use and environmental relevance

Case Study 1: Product System

Beyond Meat's Beyond Burger Life Cycle Assessment:
A detailed comparison between a plant-based and an animal-based protein source

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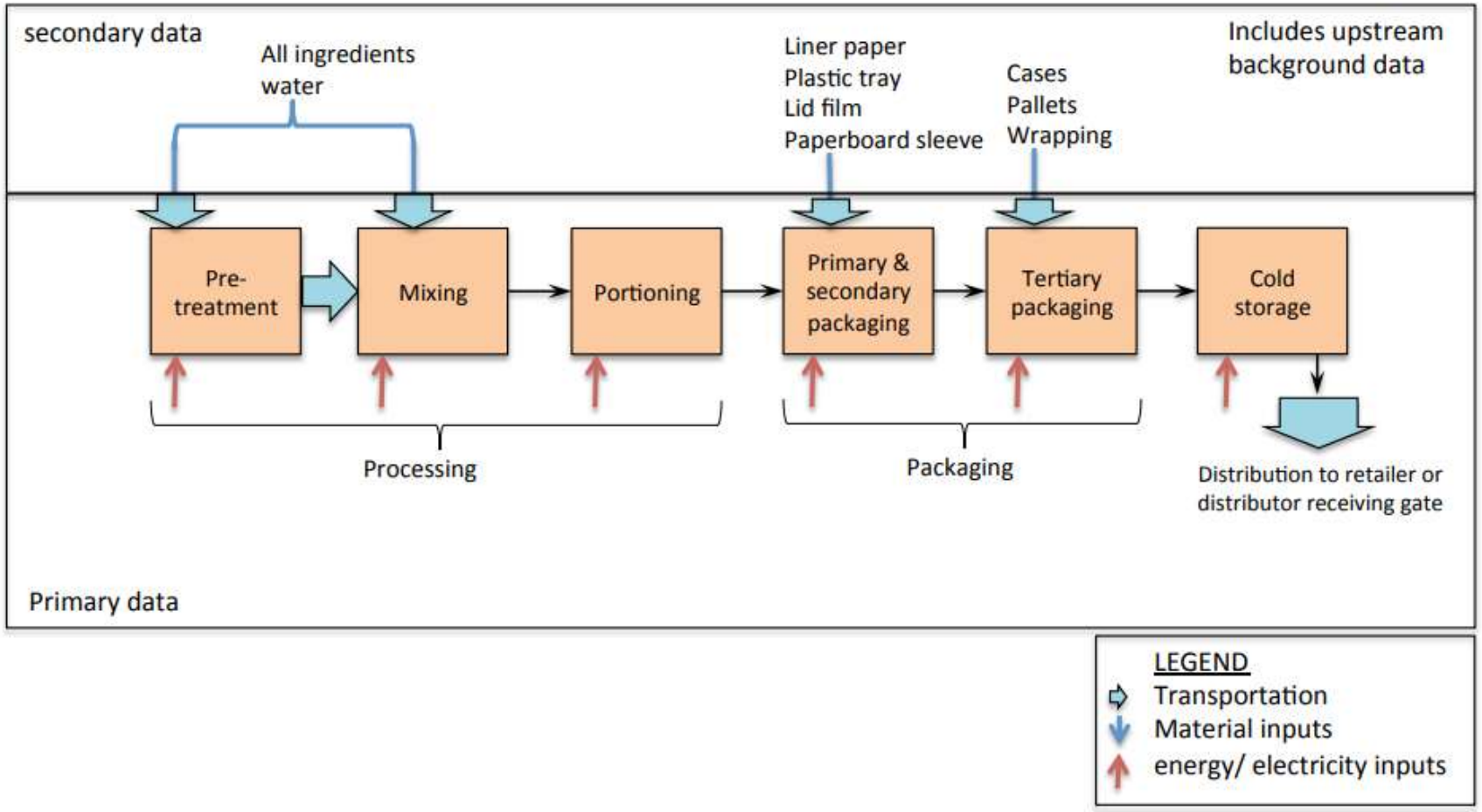

Case Study 2: Product System

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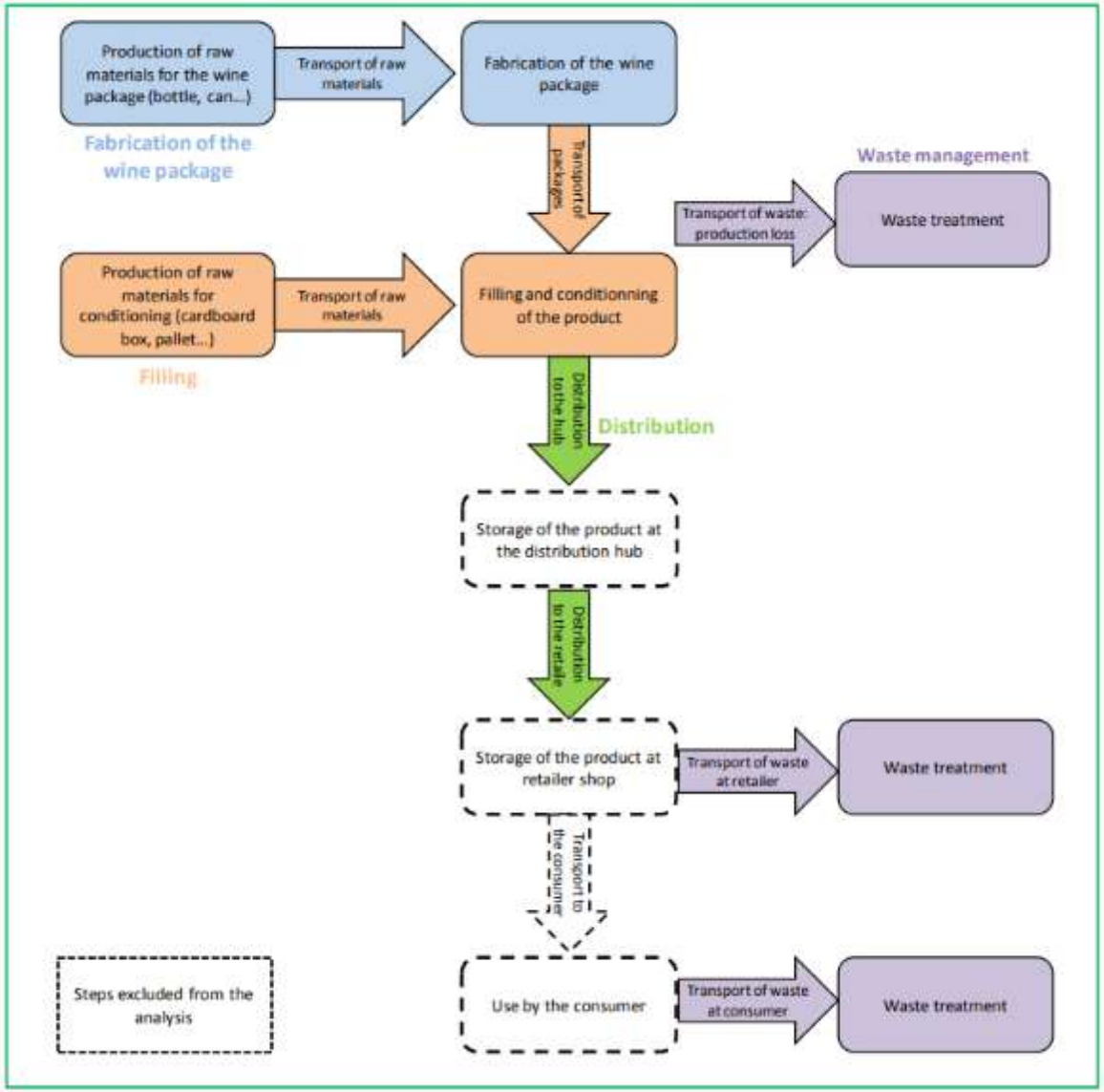

What have we learned?

- ✓ Goal & scope components of an LCA
- ✓ LCA terminology under ISO 14040/14044 for product systems and flows
- ✓ How to define system boundary, product system, functional unit and reference flows in an LCA